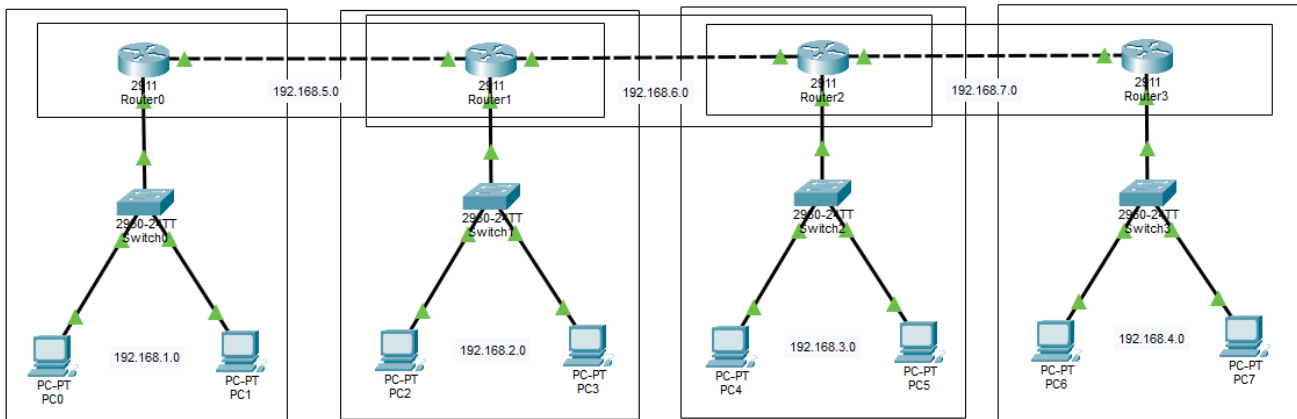


Q1. Develop Static Routing with 4 PC's, 4 Switches & 8 PC's. Make sure to highlight the Networks with their network ID's. IP Addresses should be Written in text format as well. (6 M)

Ans.



[Cisco Packet Tracer File will also be uploaded along side this PDF.]

Q2. Write the difference between Hub, Switch & Router in Tabular Format. Explain why router is the most intelligent & Hub is least intelligent device. (3 M)

Ans.

| Hub                                                                          | Switch                                                                          | Router                                                                  |
|------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| It falls in 1 <sup>st</sup> (Physical) OSI Layer.                            | It falls in 2 <sup>nd</sup> (Data link) OSI layer.                              | It falls in 3 <sup>rd</sup> (Network) OSI layer.                        |
| Its primary function is to connect multiple devices on a single LAN segment. | Its primary function is to connect devices within same Local Area Network.      | Its primary function is to connect different networks (I.e LAN to WAN). |
| It delivers data by broadcasting data to all connected ports.                | It delivers data by uni casting data directly to the specific destination port. | It delivers data by routing data packets based on IP addresses.         |
| It does not perform device identification.                                   | It uses MAC address.                                                            | It uses IP address.                                                     |
| It has no memory or lookup table.                                            | It maintains a MAC address table.                                               | It maintains a routing table.                                           |
| It uses Half Duplex connection.                                              | It uses Full Duplex connection.                                                 | It uses Full Duplex connection.                                         |

A **Hub** is considered the least intelligent because it operates at Layer 1 purely as an electrical repeater, completely unaware of MAC or IP addresses, and blindly broadcasts every incoming signal to all connected ports, causing frequent data collisions and network congestion. In contrast, a **Router** is the most intelligent because it operates at Layer 3 using advanced software logic, maintaining dynamic routing tables to calculate the optimal path for data across separate networks, while actively managing

traffic through Network Address Translation (NAT), firewall security, Quality of Service (QoS), and broadcast isolation.

Q3. What are Random Access Protocols? Why do we need them? Explain each category in detail with diagram and example. Also Explain difference between CSMA/CD & CSMA/CA. (3 M)

Ans.

#### Random Access Protocol:

Protocols in the MAC sub-layer where nodes access a shared medium dynamically without central control or scheduled time slots. Any node with data can transmit at any time.

#### Why Needed:

- **Dynamic Traffic Handling:** Prevents bandwidth wastage from fixed allocation during unpredictable, bursty data transmissions.
- **Full Channel Speed:** Allows a single active node to utilize 100% of the channel capacity when idle.
- **No Central Controller:** Eliminates master-node overhead and single points of failure.

#### Categories of Random Access Protocols

1. **ALOHA Protocol:** Transmits data immediately without checking channel status.
  1. **Pure ALOHA:** Transmits whenever data is ready. Collisions cause random backoff. (*Max Efficiency: ~18.4%*)
  2. **Slotted ALOHA:** Time is divided into discrete slots. Nodes can only transmit at the **start** of a slot. (*Max Efficiency: ~36.8%*)
  3. **Example:** Satellite communication & 4G/5G initial random access channels.
2. **CSMA:** Nodes "listen before talking" to reduce collisions.
  1. **Types:**
    1. **1-Persistent:** Senses continuously; transmits immediately when idle.
    2. **Non-Persistent:** If busy, waits a random delay before sensing again.
    3. **p-Persistent:** Used in slotted media; transmits with probability  $p$  when idle.
  2. **Example:** Legacy LAN broadcast channels.
3. **CSMA/CD:** Nodes "listen while talking." If two signals overlap, a voltage spike is detected, transmission stops mid-frame, and a jamming signal is sent before executing Binary Exponential Backoff.
  1. **Example:** Traditional Wired Ethernet.
4. **CSMA/CA:** Used where nodes cannot detect collisions mid-transmission (e.g., wireless signal fading). Uses IFS (Inter-Frame Space), RTS/CTS (Request/Clear to Send) handshaking, and explicit ACKs to prevent collisions before sending.
  1. **Example:** Wi-Fi.

#### Difference between CSMA/CD & CSMA/CA:

| CSMA/CD | CSMA/CA |
|---------|---------|
|---------|---------|

|                                                  |                                                              |
|--------------------------------------------------|--------------------------------------------------------------|
| Its full form is Collision Detection.            | Its full form is Collision Avoidance                         |
| Its Wired (Ethernet)                             | Its Wireless (Wifi)                                          |
| It aborts transmission upon collision detection. | It prevents collisions via RTS/CTS reservation & IFS timers. |
| It Detects voltage spikes while transmitting.    | It informs nearby nodes to defer transmission.               |